

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 27

Class Property

Following theorem shows that recurrence and transience are class properties.

Theorem: Let C be a communicating class. Then either all states in C are recurrent or all are transient.

Class Property

Theorem: Let C be a communicating class. Then either all states in C are recurrent or all are transient.

Proof:

- Consider any pair of states $i, j \in C$, $i \neq j$. Then there exist $n, m \geq 0$ such that $p_{ij}^{(n)} > 0$ and $p_{ji}^{(m)} > 0$.
- Now, suppose that i is transient.
- For any $r \geq 0$,

$$p_{ii}^{(n+r+m)} \geq p_{ij}^{(n)} p_{jj}^{(r)} p_{ji}^{(m)}.$$

- So,

$$\sum_{r=0}^{\infty} p_{jj}^{(r)} \leq \frac{1}{p_{ij}^{(n)} p_{ji}^{(m)}} \sum_{r=0}^{\infty} p_{ii}^{(n+r+m)} < \infty.$$

- Hence, j is also transient.

Class Property

Theorem: Let C be a communicating class. Then either all states in C are recurrent or all are transient.

Proof:

- Suppose that i is recurrent.
- If possible, assume that j is transient.
- As $j \leftrightarrow i$, i is transient, which is a contradiction.
- Therefore, j is a recurrent state.

Remark: A communicating class is called recurrent (transient) class if all the states of the class are recurrent (transient).

Remark: If a MC is irreducible, then all its states are either recurrent or transient. Accordingly, we say that the MC is recurrent or transient.

Class Property

Theorem: Every recurrent class is closed.

Proof:

- Suppose that C is a recurrent class.
- If possible, suppose that C is not closed. Then there exists $i \in C, j \notin C$ and $k \geq 1$ such that

$$P_i(X_k = j) > 0.$$

- Since,

$$P_i(\{X_k = j\} \cap \{X_n = i \text{ for infinitely many } n\}) = 0,$$

we must have

$$P_i(X_n = i \text{ for infinitely many } n) < 1.$$

- Hence, i is not recurrent and thereby C is not recurrent, which is a contradiction.

Class Property

The following theorem is partial converse of the previous theorem.

Theorem: Every finite closed class is recurrent.

Proof:

- Suppose that C is closed and finite and $\{X_n : n \geq 0\}$ starts in C .
- Then for some $i \in C$, we must have

$$\begin{aligned} 0 &< P(X_n = i \text{ for infinitely many } n) \\ &= P(X_n = i \text{ for some } n) P_i(X_n = i \text{ for infinitely many } n). \end{aligned}$$

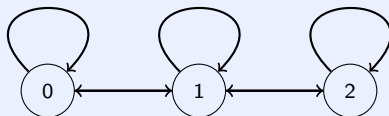
- Hence,

$$P_i(X_n = i \text{ for infinitely many } n) > 0.$$

- Thus, i is not transient and thereby has to be recurrent.
- Since, recurrence is a class property, C is recurrent.

Example

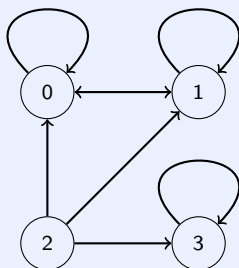
Example 1: Recall Example 1 from Lecture 23.



- The MC has a single closed class $\{0, 1, 2\}$.
- The class $\{0, 1, 2\}$ is finite.
- This is a recurrent MC.

Example

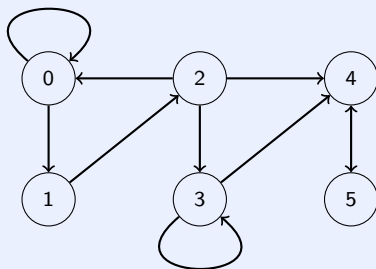
Example 2: Recall Example 2 from Lecture 23.



- $\{0, 1\}$, $\{2\}$ and $\{3\}$ are communicating classes.
- $\{0, 1\}$ and $\{3\}$ are closed and finite.
- $\{2\}$ is not closed.
- The classes $\{0, 1\}$ and $\{3\}$ recurrent.
- The class $\{2\}$ is transient.

Example

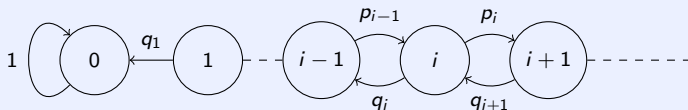
Example 3: Recall Example 3 from Lecture 23.



- $\{0, 1, 2\}$, $\{3\}$ and $\{4, 5\}$ are communicating classes.
- $\{4, 5\}$ is closed and finite.
- The classes $\{0, 1, 2\}$ and $\{3\}$ are not closed.
- The classes $\{0, 1, 2\}$ and $\{3\}$ are transient.
- Class $\{4, 5\}$ is recurrent.

Example

Example 4: [Birth-and-death Model] Consider the following MC:



- There are two communicating classes, *viz.*, $\{0\}$ and $\{1, 2, 3, \dots\}$.
- The class $\{0\}$ is closed and finite, hence recurrent.
- The class $\{1, 2, 3, \dots\}$ is not closed, and hence transient.